



Student Number

**YEAR 12 ASSESSMENT BLOCK
SEMESTER 1**

Mathematics Extension 2

1. A○ B○ C○ D○
 2. A○ B○ C○ D○
 3. A○ B○ C○ D○
 4. A○ B○ C○ D○
 5. A○ B○ C○ D○
 6. A○ B○ C○ D○
 7. A○ B○ C○ D○
 8. A○ B○ C○ D○



Barker
College

--	--	--	--	--	--	--	--	--

Student Number

2022

**YEAR 12
ASSESSMENT BLOCK**

Mathematics Extension 2

Staff Involved:

GDH KJL
ARM*

**12:00 NOON
FRIDAY 1 APRIL**

40 copies

TOPICS COVERED:

Complex Numbers I & II

Proof

**General
Instructions**

- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided separately
- For questions in Section II, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for careless or poorly arranged working
- Diagrams are not to scale
- Write your student number at the top of this page

**Total marks:
80**

Section I – 8 marks (pages 4-6)

- Attempt Questions 1-8
- Allow about 15 minutes for this section

Section II – 72 marks (pages 7-12)

- Attempt Questions 9-14
- Allow about 1 hour and 45 minutes for this section

Section I

8 marks

Attempt Questions 1–8

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–8.

1 For real x , which one of these true statements has a converse that is also true?

A. If x is positive, then x^2 is positive.

B. If $x > 3$, then $x - 3 \geq 0$.

C. If $x > 0$, then $-x < 0$.

D. If x is divisible by four then x is even.

2 What is the value of $i + i^2 + i^3 + i^4 + \dots + i^{2022}$?

A. 0

B. i

C. $1 + i$

D. $-1 + i$

3 How many complex numbers satisfy $|z - 1| = |z + 1| = |z - i|$?

- A. 0
- B. 1
- C. 2
- D. Infinitely many

4 What is the area of the triangle on the Argand diagram with vertices at z , iz and $z + iz$?

- A. $|z|^2$
- B. $\frac{1}{2}|z|^2$
- C. $\frac{\sqrt{3}}{2}|z|^2$
- D. $\frac{3}{2}|z|^2$

5 Which one of the following statements is false?

- A. $\forall a, b \in \mathbb{R}, \sqrt{ab} \in \mathbb{C}$
- B. $\forall a \in \mathbb{R}, \exists b \in \mathbb{R}: \sqrt{ab} \in \mathbb{R}$
- C. $\forall a \in \mathbb{R}, \exists b \in \mathbb{R}: \frac{a}{b} \in \mathbb{C}$
- D. $\forall a, b \in \mathbb{R}, \frac{a}{b} \in \mathbb{R}$

6 What is the smallest positive integer, k , for which $(1 + i)^{2k} = (1 - i)^{2k}$?

- A. 1
- B. 2
- C. 3
- D. 4

7 Which expression is equivalent to $\left(\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}\right) \frac{\sqrt{3} + i}{i - 1}$?

- A. $\sqrt{2}e^{i\frac{13\pi}{12}}$
- B. $\sqrt{2}e^{i\frac{11\pi}{12}}$
- C. $\sqrt{2}e^{i\frac{7\pi}{12}}$
- D. $\sqrt{2}e^{i\frac{5\pi}{12}}$

8 If the cube roots of unity are 1, w and w^2 , what are the roots of $(z - 1)^3 + 8 = 0$?

- A. $z = -1, -1, -1$
- B. $z = -1, -1 + 2w, -1 - 2w^2$
- C. $z = -1, 1 + 2w, 1 - 2w^2$
- D. $z = -1, 1 - 2w, 1 - 2w^2$

End of Section I

Section II

72 marks

Attempt Questions 9–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

For questions in Section II, your responses should include relevant mathematical reasoning and/or calculations.

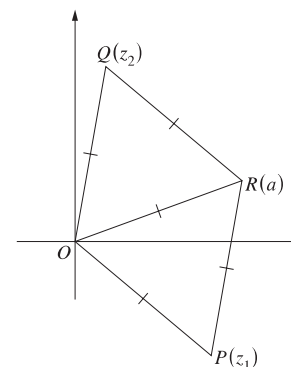
Question 9 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the complex numbers $z = 12e^{i\frac{\pi}{6}}$ and $w = 3e^{i\frac{\pi}{4}}$.
- (i) Express z in the simplest form of $x + iy$. 1
- (ii) Express $\frac{z}{w}$ in the simplest form of $re^{i\theta}$. 1
- (iii) Express $\sqrt{zw}$ in the simplest form of $re^{i\theta}$. 2
- (b) (i) Find the two square roots of $-3 + 4i$. 2
- (ii) Hence, solve $z^2 - z + 1 - i = 0$. 2
- (c) (i) Express $\sqrt{3} - i$ in modulus-argument form. 1
- (ii) Express $(\sqrt{3} - i)^7$ in modulus-argument form. 1
- (iii) Hence, express $(\sqrt{3} - i)^7$ in the simplest form of $x + iy$. 2

Question 10 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the statement, S : if $3a^2 - 4a + 5$ is even, then a is odd.
- (i) State the contrapositive of S . 1
- (ii) Use the contrapositive to prove S . 2
- (b) (i) Sketch the region on the Argand diagram satisfying both the inequalities $|z - \bar{z}| < 2$ and $|z - 1| \geq 1$. 4
- (ii) Consider the arguments of all the complex numbers in the region from (i). List all the values in $(-\pi, \pi]$ that are not included. 1

(c)



The points P , Q and R on the Argand diagram represent the complex numbers z_1 , z_2 and a , respectively.

The triangles OPR and OQR are equilateral with unit sides so that $|z_1| = |z_2| = |a| = 1$.

Let $w = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$.

- (i) Explain why $z_2 = aw$. 1
- (ii) Show that $z_1 z_2 = a^2$. 1
- (iii) Show that z_1 and z_2 are the roots of $z^2 - az + a^2 = 0$. 2

Question 11 (12 marks) Use a SEPARATE writing booklet.

- (a) The polynomial $P(z) = z^3 - 5z^2 + az + b$, where a and b are real, has one root at $z = 3 - 2\sqrt{2}i$. Find the values of a and b . 3
- (b) (i) Prove that $\frac{1}{10} - \frac{1}{11} < \frac{1}{100}$. 1
- (ii) For positive real n , prove that $\frac{1}{n} - \frac{1}{n+1} < \frac{1}{n^2}$. 2
- (c) The complex number z moves such that $\operatorname{Im}\left(\frac{1}{z-i}\right) = 2$. 3
- Show that the locus of z is a circle and find its centre and radius.
- (d) Prove by induction that $x^n - y^n$ is divisible by $x^2 - y^2$ for all positive even integers n , where x and y are integers. 3

Question 12 (12 marks) Use a SEPARATE writing booklet.

- (a) (i) Show that $a^2 + b^2 > 2ab$, where a and b are distinct positive real numbers. 1
- (ii) Hence, show that $a^2 + b^2 + c^2 > ab + bc + ca$, where a , b and c are distinct positive real numbers. 1
- (iii) Hence, or otherwise, prove that 2
- $$\frac{a^2b^2 + b^2c^2 + c^2a^2}{a + b + c} > abc$$
- where a , b and c are distinct positive real numbers.
- (b) Prove that n^2 is divisible by 5 **if and only if** n is divisible by 5. 3
- (c) (i) Show that $(1 + i \tan \theta)^n - (1 - i \tan \theta)^n = \frac{2 \sin n\theta}{\cos^n \theta} i$ 2
- where $\cos \theta \neq 0$ and n is a positive integer.
- (ii) Use the result from (i) to solve $(1 + z)^4 - (1 - z)^4 = 0$, where $\operatorname{Re}(z) = 0$. 2
- (iii) Briefly explain why $(1 + z)^4 - (1 - z)^4 = 0$ only has three roots, not four. 1

Question 13 (12 marks) Use a SEPARATE writing booklet.

- (a) Consider the equation $z^5 + 1 = 0$.
- (i) Draw a clearly labelled sketch of the roots of $z^5 + 1 = 0$ on an Argand diagram. **1**
- (ii) Factorise $z^5 + 1$ into a product of linear and quadratic factors with real coefficients. **2**
- (iii) Show that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} = \frac{1}{2}$ and $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = \frac{-1}{4}$. **3**
- (iv) Write a quadratic equation with integer coefficients that has roots $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$. **2**
- Hence find the value of $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$ as surds.

- (b) Prove by induction for integers, $n \geq 2$ that **4**

$$1^3 + 2^3 + \dots + (n-1)^3 < \frac{n^4}{4} < 1^3 + 2^3 + \dots + n^3$$

Question 14 (12 marks) Use a SEPARATE writing booklet.

- (a) Use proof by contradiction to prove the following statement: **3**
- If a is rational and b is irrational, then $a + b$ is irrational.
- (b) By proving the function $f(x) = x - \ln \left(1 + x + \frac{1}{2}x^2 \right)$ is increasing for $x \neq 0$, **4**
- show that $e^x < 1 + x + \frac{1}{2}x^2$ for $x < 0$.
- (c) (i) If $z = \cos \theta + i \sin \theta$, express $z^n + z^{-n}$ in terms of θ . **1**
- (ii) Consider the following function where k is real: **3**

$$f(\theta) = 1 + k \cos \theta + k^2 \cos 2\theta + k^3 \cos 3\theta + \dots + k^n \cos n\theta + \dots$$

Using the result in (i) and by expressing $2f(\theta)$ as the sum of two geometric series, prove that

$$f(\theta) = \frac{1 - k \cos \theta}{1 - 2k \cos \theta + k^2} \quad \text{for } |k| < 1$$

You may assume that complex numbers can be used in geometric series calculations, just as real numbers can.

- (iii) Verify the result in (ii) for $|k| < 1$ and $\theta = \frac{\pi}{2}$. **1**

End of paper

Extension 2 Mathematics

Solutions to 1st April, 2022.

Q1, The converse of $p \Rightarrow q$ is $q \Rightarrow p$.

If $-x < 0$ then $x > 0$ is true.

Hence (C)

Q2, Now $i + i^2 + i^3 + i^4$

$$= i - 1 - i + 1$$

$$= 0$$

So every 4 terms sums to 0.

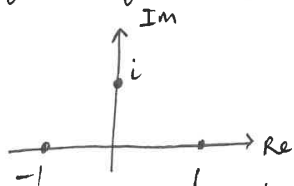
And $4 \overline{) 2022}$ remainder 2.

$$\text{So sum} = 0 + i + i^2$$

$$= -1 + i$$

Hence (d)

Q3, $|z-1| = |z+1| = |z-i|$



The only complex number equidistant from -1, 1 and i is the origin.

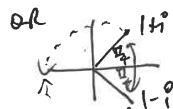
Hence (b)

OR: $(1+i)^2 = 2i$ $(1-i)^2 = -2i$

$$\therefore (2i)^k = (2i)^k$$

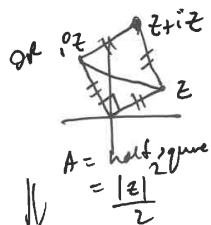
$$(2i)^k = (2i)^k (-1)^k$$

$$\therefore (-1)^k = 1 \therefore k=2$$



$$(1+i)^4 = (1-i)^4 = (\sqrt{2})^4 \cos \pi = -4$$

$$\therefore k=2$$

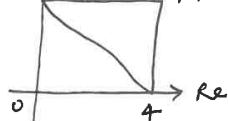


Q4, Let $z = 4$

$$iz = 4i$$

$$z + iz = 4 + 4i$$

Then



Area of $\Delta = 8 \text{ units}^2$

$$8 = \frac{1}{2} (\sqrt{16})^2$$

$$A = \frac{1}{2} |z|^2$$

Hence (b)

Q5, The statement

$$\forall a, b \in \mathbb{R}, \frac{a}{b} \in \mathbb{R}$$

is false when $b = 0$.

Hence (d)

Q6, when $k=2$,

$$(1+i)^4 = 1 + 4i + 6i^2 + 4i^3 + i^4 = -4$$

$$(1-i)^4 = 1 - 4i + 6i^2 - 4i^3 + i^4 = -4$$

Hence (b)

$$Q7, (\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}) \times \frac{\sqrt{3}+i}{-1+i}$$

$$= \text{cis}(-\frac{\pi}{3}) \times \left(\frac{2 \text{cis} \frac{\pi}{6}}{\sqrt{2} \text{cis} \frac{3\pi}{4}} \right)$$

$$= \text{cis}(-\frac{\pi}{3}) \times \sqrt{2} \text{cis}(-\frac{7\pi}{12})$$

$$= \sqrt{2} \text{cis}(-\frac{11\pi}{12})$$

$$= \sqrt{2} \text{cis}(13\pi/12)$$

Hence (A)

Q8, Test roots by substitution.

$$\text{When } z = 1-2w$$

$$(z-1)^3 + 8 = 0 \text{ becomes}$$

$$(1-2w-1)^3 + 8 = 0$$

$$-8w^3 + 8 = 0$$

$$\text{but } w^3 = 1$$

$$\therefore -8 + 8 = 0$$

Hence $z = 1-2w$ is a root

When $z = 1-2w^2$ is substituted we get

$$(1-2w^2-1)^3 + 8$$

$$= -8w^6 + 8$$

$$= -8 + 8$$

$$= 0$$

Hence $z = 1-2w^2$ is also a root. Hence (d)

OR Roots of unity:

$$w = \text{cis} \frac{2\pi}{3}, w^2 = \text{cis} \frac{-2\pi}{3}$$

$$\text{So } 2w = 2 \text{cis} \frac{2\pi}{3}$$

$$1-2w = -2 \text{cis} \frac{2\pi}{3} = 2 \text{cis} \frac{\pi}{3}$$

$$2w^2 = 2 \text{cis} \frac{-2\pi}{3} = 2 \text{cis} \frac{\pi}{3}$$

$$1-2w^2 = -2 \text{cis} \frac{2\pi}{3} = 2 \text{cis} \frac{\pi}{3}$$

$$\text{So for } (z-1)^3 = -8$$

$$(z-1)^3 = 8 \text{cis}(\pi + 2k\pi)$$

$$z-1 = 2 \text{cis} \frac{\pi}{3}, 2 \text{cis} \pi, 2 \text{cis} \frac{5\pi}{3}$$

$$= -2w^2, -2, -2w$$

$$\therefore z = 1-2w^2, 1, 1-2w$$

Q9,

$$(a) z = 12 e^{i \frac{\pi}{6}}$$

$$w = 3 e^{i \frac{\pi}{4}}$$

$$(i) \frac{z}{w} = \frac{12 e^{i \frac{\pi}{6}}}{3 e^{i \frac{\pi}{4}}} = 4 e^{i \frac{-\pi}{12}}$$

$$(ii) \sqrt{zw} = \sqrt{36 e^{i \frac{5\pi}{12}}} = \pm 6 e^{i \frac{5\pi}{24}}$$

$$(b) (i) \text{ let } (x+iy)^2 = -3+4i \\ \text{then } x^2-y^2 = -3 \text{ and } 2xyi = 4i \\ y = \frac{2}{x}$$

$$x^2 - \left(\frac{2}{x}\right)^2 = -3$$

$$x^4 + 3x^2 - 4 = 0$$

$$(x^2+4)(x^2-1) = 0 \\ x = \pm 1$$

$$\text{Hence } \pm(1+2i)$$

$$(ii) z^2 - z + 1 - i = 0 \\ z = \frac{1 \pm \sqrt{-3+4i}}{2} \\ z = \frac{1 \pm (1+2i)}{2} \\ z = -i, 1+i$$

$$(c) (i) \sqrt{3} - i = 2 \operatorname{cis} \left(\frac{-\pi}{6} \right)$$



$$(ii) (\sqrt{3}-i)^7 = (2 \operatorname{cis} \frac{-\pi}{6})^7 = 128 \operatorname{cis} \left(\frac{-7\pi}{6} \right) = 128 \operatorname{cis} \left(\frac{5\pi}{6} \right)$$

$$(iii) (\sqrt{3}-i)^7 = 128 \left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6} \right) = 128 \left(-\frac{\sqrt{3}}{2} + i \frac{1}{2} \right) = -64\sqrt{3} + 64i$$

$$Q10, (a) S: \text{ if } 3a^2-4a+5 \text{ is even, then } a \text{ is odd.}$$

(i) The contrapositive of S is:

"If a is even, then $3a^2-4a+5$ is odd."

(ii) Suppose that a is even.

Let $a = 2j$, j an integer.

$$\text{Then } 3a^2-4a+5 = 3(2j)^2-4(2j)+5 = 12j^2-8j+5 = 2[6j^2-4j]+5 \quad \text{--- ①}$$

which is an even number.

In ① an even number added to an odd number makes an odd number.

Hence, if a is even, then $3a^2-4a+5$ is odd.

Therefore, if $3a^2-4a+5$ is even, then a is odd.

Q10, (b)

(i) $|z - \bar{z}| < 2$

$$|x + iy - (x - iy)| < 2$$

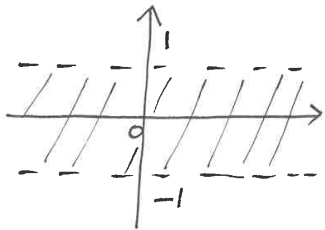
$$|2iy| < 2$$

$$\sqrt{4y^2} < 2$$

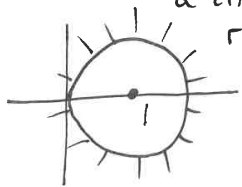
$$\pm 2y < 2$$

$$-1 < y < 1$$

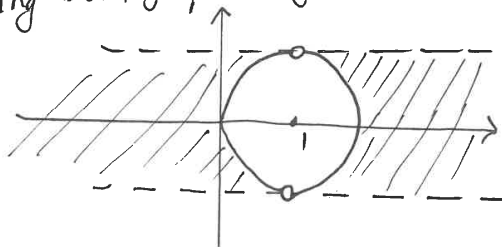
gives



and $|z - 1| \geq 1$ gives
a circle, centre (1, 0)
radius 1 unit



Putting both graphs together gives:



(ii) $\frac{\pi}{4}$ and $-\frac{\pi}{4}$ are the only two values of the argument of z not representing any complex number in the shaded region. These values correspond to the "holes" in the circle.

(c) (i) Now $\angle QOR = \frac{\pi}{3}$
we can rotate OR anticlockwise about O by multiplying a by $\text{cis}(\frac{\pi}{3})$.

$$\begin{aligned} \text{Hence } z_2 &= a \times (\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}) \\ &= a\omega \\ &\text{as required.} \end{aligned}$$

(ii) Now OP is OR rotated $\frac{\pi}{3}$ clockwise. This can be done by dividing a by $\text{cis} \frac{\pi}{3}$ or multiplying by $\text{cis}(-\frac{\pi}{3})$.

$$\begin{aligned} \text{So } z_1 &= a \text{cis}(-\frac{\pi}{3}) \\ \therefore z_1 z_2 &= a \text{cis}(-\frac{\pi}{3}) \times a \text{cis}(\frac{\pi}{3}) \\ &= a^2 \text{cis}(0) \\ &= a^2 \\ &\text{as required.} \end{aligned}$$

(iii) Now $z_1 = a \text{cis}(-\frac{\pi}{3})$
 $= a(\frac{1}{2} - \frac{\sqrt{3}}{2}i)$

and $z_2 = a \text{cis}(\frac{\pi}{3})$
 $= a(\frac{1}{2} + \frac{\sqrt{3}}{2}i)$

Now sum of roots above is

$$\frac{1}{2}a + \frac{1}{2}a = a \quad \leftarrow \text{same!}$$

And sum of roots of equation is

$$-\frac{b}{a} = \frac{-(-a)}{1} = a \quad \leftarrow \text{same!}$$

The product of the roots above is

$$a^2(\frac{1}{2} + \frac{\sqrt{3}}{2}i)(\frac{1}{2} - \frac{\sqrt{3}}{2}i)$$

$$= a^2(1)$$

$$= a^2 \quad \leftarrow \text{same!}$$

And product of roots of equation is

$$\frac{c}{a} = \frac{a^2}{1} = a^2 \quad \leftarrow \text{same!}$$

$\therefore z_1$ and z_2 are the roots of

$$z^2 - az + a^2 = 0$$

Q11// Since $3-2\sqrt{2}i$ is a root, and all the coefficients of the equation are real, then $3+2\sqrt{2}i$ is also a root.

Then the sum of the roots is

$$(3-2\sqrt{2}i) + (3+2\sqrt{2}i) + \gamma = 5$$

$$\gamma = -1$$

The product of the roots is

$$(-1) \times (3-2\sqrt{2}i) \times (3+2\sqrt{2}i)$$

$$= -17$$

$$\text{Hence } -b = -17 \Rightarrow b = 17$$

$$\therefore P(z) = z^3 - 5z^2 + az + 17$$

$$\text{but } P(-1) = (-1)^3 - 5(-1)^2 - a + 17 = 0$$

$$0 = 11 - a \Rightarrow a = 11$$

$$(b) \quad (i) \quad \frac{1}{10} - \frac{1}{11}$$

$$= \frac{11}{110} - \frac{10}{110}$$

$$= \frac{1}{110}$$

$$< \frac{1}{100}$$

(ii) To prove that $\frac{1}{n} - \frac{1}{n+1} < \frac{1}{n^2}$:

consider RHS - LHS

$$= \frac{1}{n^2} - \frac{1}{n} + \frac{1}{n+1}$$

$$= \frac{1}{n^2} - \frac{1}{n(n+1)}$$

$$= \frac{(n+1) - n}{n^2(n+1)}$$

$$= \frac{1}{n^2(n+1)} > 0$$

$\therefore \text{RHS} > \text{LHS}$

$$\therefore \frac{1}{n} - \frac{1}{n+1} < \frac{1}{n^2} \quad \text{as Required.}$$

$$\text{Q11// (c) } \operatorname{Im} \left(\frac{1}{\bar{z} - i} \right) = 2$$

let $z = x + iy$ then

$$\operatorname{Im} \left(\frac{1}{x - iy - i} \right) = 2$$

$$\operatorname{Im} \left(\frac{1}{x - i(y+1)} \times \frac{x + i(y+1)}{x + i(y+1)} \right) = 2$$

$$\operatorname{Im} \left(\frac{x + i(y+1)}{x^2 + (y+1)^2} \right) = 2$$

$$\frac{y+1}{x^2 + (y+1)^2} = 2$$

$$y+1 = 2x^2 + 2y^2 + 4y + 2$$

$$0 = 2x^2 + 2y^2 + 3y + 1$$

$$0 = x^2 + y^2 + \frac{3}{2}y + \frac{1}{2}$$

$$0 = x^2 + y^2 + \frac{3}{2}y + \frac{9}{16} + \frac{1}{2} - \frac{9}{16}$$

$$0 = x^2 + \left(y + \frac{3}{4}\right)^2 - \frac{1}{16}$$

$$\frac{1}{16} = x^2 + \left(y + \frac{3}{4}\right)^2$$

which is a circle, centre $(0, -\frac{3}{4})$ and

Radius $\frac{1}{4}$ units,

with a hole at $(0, -1)$.

Q11, (d)

When $n=2$, $x^n - y^n = x^2 - y^2$, which is divisible by $x^2 - y^2$.

Hence the statement is true when $n=2$.

Assume that the statement is true when $n=2k$, k an integer.

Assume that $x^{2k} - y^{2k}$ is div. by $x^2 - y^2$, i.e., assume that

$$x^{2k} - y^{2k} = M \cdot (x^2 - y^2) \text{ where } M \text{ is an integer.}$$

Next, try to prove that the statement is true when $n=2k+2$

$$\text{Now } x^{2k+2} - y^{2k+2}$$

$$= x^{2k} \cdot x^2 - y^{2k+2}$$

$$= (y^{2k} + M(x^2 - y^2)) \cdot x^2 - y^{2k+2} \quad (\text{by assumption})$$

$$= x^2 y^{2k} + M x^2 (x^2 - y^2) - y^{2k+2}$$

$$= y^{2k} (x^2 - y^2) + M x^2 (x^2 - y^2)$$

$$= (x^2 - y^2) [M x^2 + y^{2k}]$$

which is divisible by $(x^2 - y^2)$

as $[M x^2 + y^{2k}]$ is an integer.

Hence the statement is true by mathematical induction.

Q12, (a)

(i) Now $(a-b)^2 > 0$ if $a \neq b$

$$a^2 - 2ab + b^2 > 0$$

$$\text{Hence } a^2 + b^2 > 2ab \quad \text{--- (1)}$$

where a and b are distinct positive real numbers.

$$\text{(ii) Similarly, } a^2 + c^2 > 2ac \quad \text{--- (2)}$$

$$\text{and } b^2 + c^2 > 2bc \quad \text{--- (3)}$$

Adding (1), (2) and (3) gives

$$a^2 + b^2 + a^2 + c^2 + b^2 + c^2 > 2ab + 2ac + 2bc$$

$$2(a^2 + b^2 + c^2) > 2(ab + ac + bc)$$

$$\text{Hence } a^2 + b^2 + c^2 > ab + ac + bc, \text{ as required.}$$

$$\begin{array}{l} \text{(iii) Hence } a^2 b^2 + b^2 c^2 > 2ab^2 c \\ \text{and } b^2 c^2 + a^2 c^2 > 2abc^2 \\ \text{and } a^2 b^2 + a^2 c^2 > 2a^2 bc \end{array} \left. \vphantom{\begin{array}{l} a^2 b^2 + b^2 c^2 > 2ab^2 c \\ b^2 c^2 + a^2 c^2 > 2abc^2 \\ a^2 b^2 + a^2 c^2 > 2a^2 bc \end{array}} \right\} \begin{array}{l} \text{add together} \\ \text{to get} \end{array}$$

$$2(a^2 b^2 + b^2 c^2 + a^2 c^2) > 2(a^2 bc + ab^2 c + abc^2)$$

$$\text{so } a^2 b^2 + b^2 c^2 + a^2 c^2 > abc(a + b + c)$$

$$\text{Hence } \frac{a^2 b^2 + b^2 c^2 + a^2 c^2}{a + b + c} > abc, \text{ as required.}$$

Q12, (b)

To prove that n^2 is divisible by 5 if and only if n is divisible by 5.

We have to show

(i) That if n is divisible by 5, then n^2 is divisible by 5.

Suppose that n is divisible by 5. Let $n = 5a$.
Then $n^2 = (5a)^2 = 25a^2 = 5(5a^2)$, which is div. by 5.

(ii) That if n^2 is div by 5, then n is div. by 5.

We can prove (ii) by contrapositive.

I will show that if n is not div. by 5, then n^2 is not div. by 5.

Suppose that n is not div. by 5.

Then n leaves a remainder of $\{1, 2, 3, 4\}$ upon division by 5.

In that case, n^2 will leave a remainder of $\{1, 4, 4, 1\}$.

Hence n^2 is not divisible by 5.

So, if n is not divisible by 5, then n^2 is not div. by 5.

So, if n^2 is div. by 5, then n is div. by 5.

c) (i) To show that

$$(1 + i \tan \theta)^n - (1 - i \tan \theta)^n = \frac{2 \sin n\theta}{\cos^n \theta} i$$

where n is an integer and $\cos \theta \neq 0$.

$$\begin{aligned} \text{LHS} &= \left(1 + \frac{i \sin \theta}{\cos \theta}\right)^n - \left(1 - \frac{i \sin \theta}{\cos \theta}\right)^n \\ &= \left(\frac{\cos \theta + i \sin \theta}{\cos \theta}\right)^n - \left(\frac{\cos \theta - i \sin \theta}{\cos \theta}\right)^n = \frac{(\cos \theta + i \sin \theta)^n}{\cos^n \theta} - \frac{(\cos \theta - i \sin \theta)^n}{\cos^n \theta} \\ &= \frac{\cos n\theta + i \sin n\theta - (\cos n\theta - i \sin n\theta)}{\cos^n \theta} \\ &= \frac{\cos n\theta + i \sin n\theta - \cos n\theta + i \sin n\theta}{\cos^n \theta} = \frac{2i \sin n\theta}{\cos^n \theta} = \frac{2 \sin n\theta}{\cos^n \theta} i = \text{RHS} \end{aligned}$$

(ii) Now part (i) says that

$$(1 + i \tan \theta)^n - (1 - i \tan \theta)^n = \frac{2 \sin n\theta}{\cos^n \theta} i$$

Let $z = i \tan \theta$, then

$$(1 + z)^n - (1 - z)^n = \frac{2 \sin n\theta}{\cos^n \theta} i$$

And if $n = 4$, we have

$$(1 + z)^4 - (1 - z)^4 = \frac{2 \sin 4\theta}{\cos^4 \theta} i$$

Hence the solutions of

$$(1 + z)^4 - (1 - z)^4 = 0$$

are the solutions of $\frac{2 \sin 4\theta}{\cos^4 \theta} = 0$

$$\text{So } 2\sin 4\theta = 0$$

$$\sin 4\theta = 0$$

$$4\theta = 0, \pi, 2\pi, 3\pi, 4\pi$$

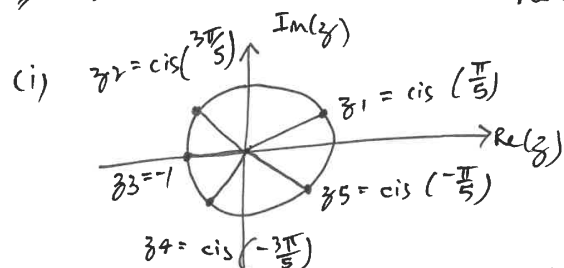
$$\theta = 0, \frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}, \pi$$

$$i \tan \theta = 0, i, 0, -i, 0$$

$$\therefore z = 0, \pm i$$

$$\begin{aligned} \text{(iii) Now } (1+z)^4 - (1-z)^4 &= ((1+z)^2 + (1-z)^2)((1+z)^2 - (1-z)^2) \\ &= 2(1+z^2)(4z) \\ &= 8z(1+z^2) \text{ which is of degree 3,} \\ &\text{hence only 3 solutions.} \end{aligned}$$

Q3 (a)



The 5 roots of $z^5 + 1 = 0$ are equally spaced around the unit circle, separated by $\frac{2\pi}{5}$ radians.

The roots are $z_1 = \text{cis}(\frac{\pi}{5})$

$$z_2 = \text{cis}(\frac{2\pi}{5})$$

$$z_3 = \text{cis}(\pi) = -1$$

$$z_4 = \text{cis}(-\frac{2\pi}{5})$$

$$z_5 = \text{cis}(-\frac{\pi}{5})$$

$$\text{(ii) } z^5 + 1 = (z+1)(z^4 - z^3 + z^2 - z + 1)$$

$$= (z+1)(z - \text{cis}(\frac{\pi}{5}))(z - \text{cis}(\frac{2\pi}{5}))(z - \text{cis}(\frac{3\pi}{5}))(z - \text{cis}(\frac{4\pi}{5}))$$

$$= (z+1)(z^2 - 2\cos\frac{\pi}{5}z + 1)(z^2 - 2\cos\frac{2\pi}{5}z + 1)$$

$$\text{(iii) Now the sum of the roots is } -\frac{b}{a} = 0.$$

$$\text{Hence } \text{cis}(\frac{\pi}{5}) + \text{cis}(\frac{2\pi}{5}) + (-1) + \text{cis}(-\frac{2\pi}{5}) + \text{cis}(-\frac{\pi}{5}) = 0$$

$$\begin{aligned} \cos(\frac{\pi}{5}) + i\sin(\frac{\pi}{5}) + \cos(\frac{2\pi}{5}) + i\sin(\frac{2\pi}{5}) + (-1) \\ + \cos(-\frac{2\pi}{5}) + i\sin(-\frac{2\pi}{5}) + \cos(-\frac{\pi}{5}) + i\sin(-\frac{\pi}{5}) = 0 \end{aligned}$$

$$\text{So } 2\cos(\frac{\pi}{5}) + 2\cos(\frac{2\pi}{5}) = 1$$

$$\text{Hence } \cos(\frac{\pi}{5}) + \cos(\frac{2\pi}{5}) = \frac{1}{2}, \text{ as required.}$$

By equating the coefficients of z^2 we have

$$0 = -2\cos\frac{\pi}{5} - 2\cos\frac{2\pi}{5} + 1 + 1 + 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

$$0 = -2(\cos\frac{\pi}{5} + \cos\frac{2\pi}{5}) + 2 + 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

$$0 = -2 \times \frac{1}{2} + 2 + 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

$$-1 = 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

$$\text{Hence } \cos\frac{\pi}{5}\cos\frac{2\pi}{5} = -\frac{1}{4}, \text{ as required.}$$

$$\text{or For } z^4 - z^3 + z^2 - z + 1 = (z^2 - 2\cos\frac{\pi}{5}z + 1)(z^2 - 2\cos\frac{2\pi}{5}z + 1)$$

$$\text{equating coefficients of } z^2: 1 = 1 + 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

$$-1 = 4\cos\frac{\pi}{5}\cos\frac{2\pi}{5}$$

(iv) A quadratic equation that has roots $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$ is

$$(x - \cos \frac{\pi}{5})(x - \cos \frac{3\pi}{5}) = 0$$

$$x^2 - x(\cos \frac{\pi}{5} + \cos \frac{3\pi}{5}) + \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = 0$$

$$x^2 - \frac{1}{2}x - \frac{1}{4} = 0, \text{ using the results from (iii)}$$

$$x = \frac{\frac{1}{2} \pm \sqrt{\frac{1}{4} - 4(-\frac{1}{4})}}{2}$$

$$x = \frac{\frac{1}{2} \pm \frac{\sqrt{5}}{2}}{2}$$

$$x = \frac{1 \pm \sqrt{5}}{4}$$

Hence $\cos(\frac{3\pi}{5}) = \frac{1 - \sqrt{5}}{4}$ because the angle is in the 2nd quadrant making $\cos$ negative

and $\cos(\frac{\pi}{5}) = \frac{1 + \sqrt{5}}{4}$ because the angle is in the 1st quadrant making $\cos$ positive.

(b)

Prove by induction for integers $n \geq 2$ that:

$$1^3 + 2^3 + \dots + (n-1)^3 < \frac{n^4}{4} < 1^3 + 2^3 + \dots + n^3$$

When $n=2$,

$$(2-1)^3 = 1^3 = 1$$

$$\frac{2^4}{4} = \frac{16}{4} = 4$$

$$1^3 + 2^3 = 8 + 1 = 9$$

and $1 < 4 < 9$.

So the statement is true when $n=2$.

Assume that the statement is true when $n=k$, i.e., assume that

$$1^3 + 2^3 + \dots + (k-1)^3 < \frac{k^4}{4} < 1^3 + 2^3 + \dots + k^3$$

Then try to prove that

$$1^3 + 2^3 + \dots + (k-1)^3 + k^3 < \frac{(k+1)^4}{4} < 1^3 + 2^3 + \dots + k^3 + (k+1)^3$$

Consider

$$\begin{aligned} & \underbrace{1^3 + 2^3 + \dots + k^3} + (k+1)^3 - \frac{(k+1)^4}{4} \\ & > \frac{k^4}{4} + (k+1)^3 - \frac{(k+1)^4}{4}, \text{ by assumption} \\ & = \frac{k^4}{4} + (k+1)^3 - \left(\frac{k^4 + 4k^3 + 6k^2 + 4k + 1}{4} \right) \\ & = (k+1)^3 - \left(k^3 + \frac{3}{2}k^2 + k + \frac{1}{4} \right) \\ & = k^3 + 3k^2 + 3k + 1 - k^3 - \frac{3}{2}k^2 - k - \frac{1}{4} \\ & = \frac{3}{2}k^2 + 2k + \frac{3}{4} \\ & > 0 \end{aligned}$$

Hence $\frac{(k+1)^4}{4} < 1^3 + 2^3 + \dots + k^3 + (k+1)^3.$

Now consider the sign of

$$\frac{(k+1)^4}{4} - [1^3 + 2^3 + \dots + (k-1)^3 + k^3]$$

$$> \frac{k^4 + 4k^3 + 6k^2 + 4k + 1}{4} - \left[\frac{k^4}{4} + k^3 \right] \text{ by the assumption}$$

$$= k^3 + \frac{3}{2}k^2 + k + \frac{1}{4} - k^3$$

$$= \frac{3}{2}k^2 + k + \frac{1}{4}, \text{ which is positive.}$$

Hence $1^3 + 2^3 + \dots + (k-1)^3 + k^3 < \frac{(k+1)^4}{4}$

and $\frac{(k+1)^4}{4} < 1^3 + 2^3 + \dots + k^3 + (k+1)^3.$

Hence $1^3 + 2^3 + \dots + k^3 < \frac{(k+1)^4}{4} < 1^3 + 2^3 + \dots + (k+1)^3,$
as required.

Q14, (a)

S: "If a is rational and b is irrational, then $a + b$ is irrational."

Proof by Contradiction:

Assume that a is rational and b is irrational and that $a + b$ is rational.

Since a and $a + b$ are rational, we can let

$$a = \frac{p}{q} \quad \text{and} \quad a + b = \frac{r}{s} \quad \text{where}$$

p and q are integers with no common factors and r and s are integers with no common factors.

$$\text{Now } a + b = \frac{r}{s}$$

$$\text{so } b = \frac{r}{s} - \frac{p}{q}$$

$$b = \frac{Rq - ps}{sq}$$

contradiction!

(This is a contradiction because $Rq - ps$ is an integer and sq is an integer, hence b is rational. But we assumed that b is irrational.)

Hence our assumption was false.

Hence if a is rational and b is irrational, then $a + b$ is irrational too.

$$(b) f(x) = x - \ln\left(1+x+\frac{1}{2}x^2\right)$$

$$f'(x) = 1 - \frac{1}{1+x+\frac{1}{2}x^2} \times (1+x)$$

$$f'(x) = \frac{1+x+\frac{1}{2}x^2-1-x}{1+x+\frac{1}{2}x^2}$$

$$f'(x) = \frac{\frac{1}{2}x^2}{1+x+\frac{1}{2}x^2} = \frac{x^2}{2+2x+x^2} = \frac{x^2}{(1+x)^2+1}$$

The numerator is positive if $x \neq 0$.

The denominator is always positive.

Hence $f'(x) > 0$ for $x < 0$.

Hence $y = f(x)$ is an increasing function for $x < 0$.

Now to show that $e^x < 1+x+\frac{1}{2}x^2$ for $x < 0$.

Notice that $f(0) = 0 - \ln(1+0+0) = 0$

and $f(x)$ is increasing for $x < 0$, so

$f(x) < 0$ for $x < 0$.

so $x - \ln\left(1+x+\frac{1}{2}x^2\right) < 0$ for $x < 0$

$$x < \ln\left(1+x+\frac{1}{2}x^2\right)$$

$\therefore e^x < 1+x+\frac{1}{2}x^2$ for $x < 0$
as required.

(c) (i)

$$z = \cos \theta + i \sin \theta$$

$$z^n = \cos(n\theta) + i \sin(n\theta) \quad \text{--- (1)}$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$z^{-n} = \cos n\theta - i \sin n\theta \quad \text{--- (2)}$$

$$\text{(1) + (2) gives } z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ = 2 \cos n\theta$$

$$(ii) f(\theta) = 1 + k \cos \theta + k^2 \cos 2\theta + k^3 \cos 3\theta + \dots + k^n \cos n\theta + \dots$$

$$2f(\theta) = 2 + k(2 \cos \theta) + k^2(2 \cos 2\theta) + k^3(2 \cos 3\theta) + \dots$$

$$= 2 + k(z + z^{-1}) + k^2(z^2 + z^{-2}) + k^3(z^3 + z^{-3}) + \dots$$

$$= [1 + k z + k^2 z^2 + k^3 z^3 + \dots] + [1 + k z^{-1} + k^2 z^{-2} + k^3 z^{-3} + \dots]$$

$$= \frac{1}{1-kz} + \frac{1}{1-kz^{-1}}$$

$$= \frac{1}{1-kz} + \frac{z}{z-k}$$

$$= \frac{1}{(1-k \cos \theta) - k i \sin \theta} + \frac{\cos \theta + i \sin \theta}{(\cos \theta - k) + i \sin \theta}$$

$$= \frac{1 - k \cos \theta + k i \sin \theta}{(1 - k \cos \theta)^2 + k^2 \sin^2 \theta} + \frac{(\cos \theta + i \sin \theta)(\cos \theta - k - i \sin \theta)}{(\cos \theta - k)^2 + \sin^2 \theta}$$

$$= \frac{1 - k \cos \theta + k i \sin \theta}{1 - 2k \cos \theta + k^2} + \frac{1 - k \cos \theta - k i \sin \theta}{1 - 2k^2 \cos \theta + k^2}$$

$$2f(\alpha) = \frac{2 - 2k \cos \alpha}{1 - 2k \cos \alpha + k^2}$$

$$\therefore f(\alpha) = \frac{1 - k \cos \alpha}{1 - 2k \cos \alpha + k^2}, \text{ as required.}$$

(iii) To verify this result, let $\alpha = \frac{\pi}{2}$ then from

$$f(\alpha) = 1 + k \cos \alpha + k^2 \cos 2\alpha + k^3 \cos 3\alpha + \dots$$

$$f\left(\frac{\pi}{2}\right) = 1 + 0 - k^2 + 0 + k^4 + 0 - k^6 + 0 + k^8 + \dots$$

$$= \frac{1}{1 - (-k^2)}$$

$$= \frac{1}{1 + k^2}$$

And when $\alpha = \frac{\pi}{2}$,

$$f(\alpha) = \frac{1 - k \cos \alpha}{1 - 2k \cos \alpha + k^2}$$

$$f\left(\frac{\pi}{2}\right) = \frac{1 - k(0)}{1 - 2k(0) + k^2}$$

$$= \frac{1}{1 + k^2}$$

Since these results are equal, the result in (ii) has been verified.

END OF SOLUTIONS